A Project Report On

Wireshark Packet Analyzer

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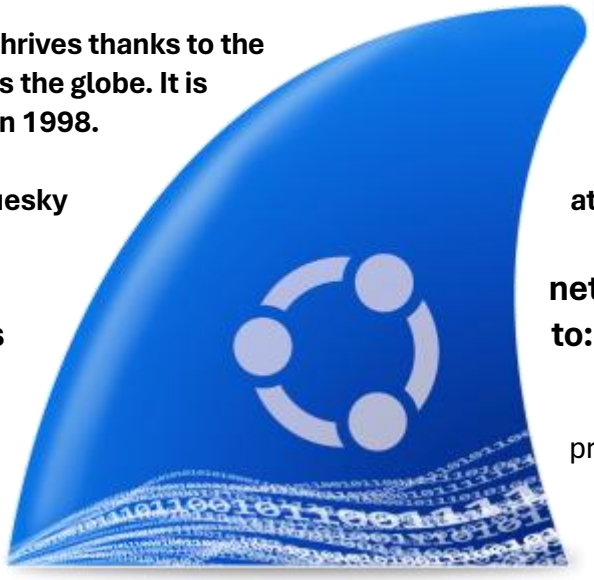
WIRESHARK

1. About Wireshark:

Wireshark is the world's foremost network protocol analyzer.
Wireshark lets you see what's happening on your network at a microscopic level. It is the de facto (and often de jure) standard across many industries and educational institutions.

Wireshark development thrives thanks to the contributions of networking experts across the globe. It is the continuation of a project that started in 1998.

You can follow on Bluesky at @wireshark.bsky.social



It helps students, network engineers, and security professionals to:

- Monitor traffic
- Debug network problems
- Analyze protocols
- Detect security issues

Wireshark was originally called Ethereal and later renamed "Wireshark".

2. Pack Sniffing:

Packet sniffing means:

A process of capturing, monitoring, and analyzing network packets that travel across a computer network. A packet is a small unit of data that contains information such as the source IP address, destination IP address, protocol type, and the actual data being transmitted.

Commonly used for:

- Network troubleshooting
- Performance monitoring
- Protocol analysis
- Detecting network errors
- Cybersecurity and threat detection

Packet sniffing helps users understand how devices communicate over the network.

Wireshark is one of the most popular packet sniffing tools. It captures live network traffic and displays each packet in a readable format so users can inspect the details.

3. Features of Wireshark:

• Deep inspection of hundreds of protocols, with more being added all the time

• Live capture and offline analysisq

• Standard three-pane packet browser

• Multi-platform: Runs on Windows, Linux, macOS, FreeBSD, NetBSD, and many others

• Captured network data can be browsed via a GUI, or via the TTY-mode TShark utility

• The most powerful display filters in the industry

• Rich VoIP analysis

• Read/write many different capture file formats: tcpdump (libpcap), Pcap NG, Catapult DCT2000, Cisco Secure IDS iplog, and many others

• Capture files compressed with gzip can be decompressed on the fly

• Live data can be read from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI, and others

• Decryption support for many protocols, including IPsec, ISAKMP, Kerberos, SNMPv3, SSL/TLS, WEP, and WPA/WPA2

• Coloring rules can be applied to the packet list for quick, intuitive analysis

• Output can be exported to XML, PostScript®, CSV, or plain text

4. Installation of Wireshark:

Steps:

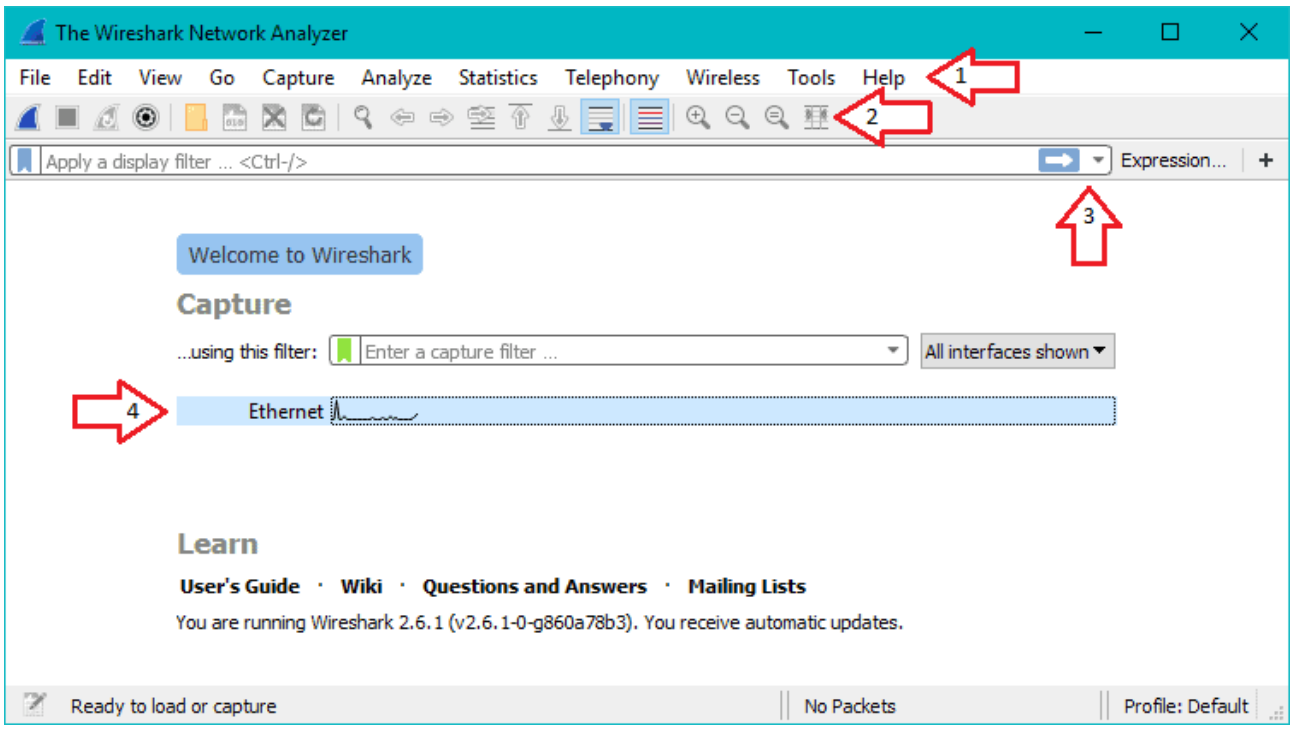
1. Go to https://www.wireshark.org/
2. Click Download
3. Install according to your OS
4. During installation, install Npcap/WinPcap
5. Launch Wireshark

Verify:

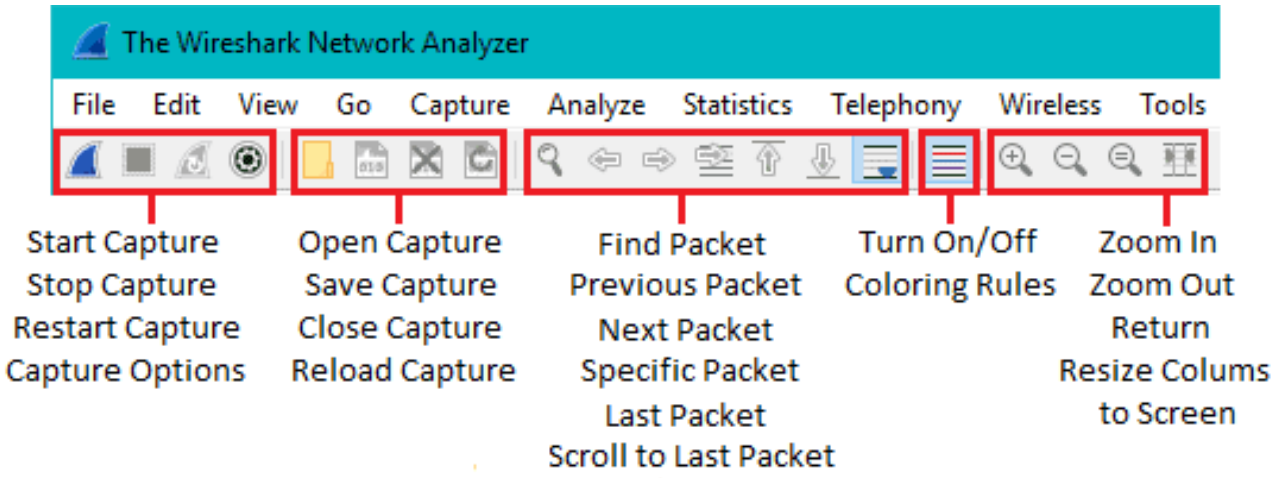
Open Wireshark → Network interfaces should be visible

5. Wireshark Interface Overview:

1. The Menu
2. The Main Toolbar
3. The Filter Toolbar
4. The Interface List



Wireshark Main Toolbar



"The Menu":

Wireshark's main menu, "The Menu," is located at the top of the window when run on Windows and Linux and the top of the screen when run on macOS.

The Menu displays 11 different items:

File:
Open/Merge capture files, save, print, export, and quit Wireshark

Edit:
Find, time reference, or mark a packet. Handle configuration profiles. Set preferences.

View:
Change display of capture data such as colorization of pakcets, showing packet in another window, zooming font, and collapsing and expanding trees.

Go:
Options to go to a specific packet

Capture:
Edit capture filters and start and stop captures.

Analyze:
Alter display filters, configure user specific decodes, enable or disable dissection of protocols, and follow TCP streams

Statistics:
Display statistic windows, summary of captured packets, protocol hierarchy stats, and more

Telephony:
Display telephony relatated stats such as media analysis, flow diagrams, protocol hierarchy stats

Wireless:
Display IEE 802.11 wireless and Bluetooth statistics

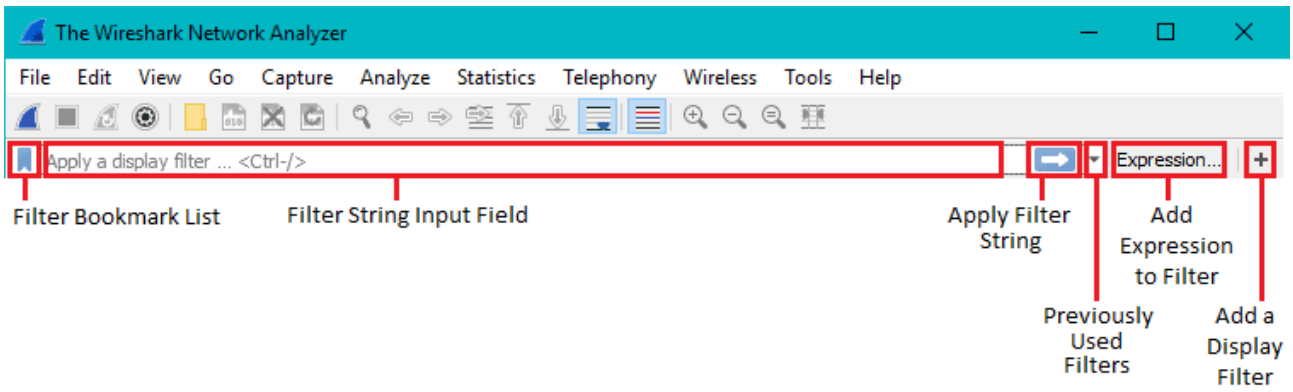
Tools:
Various tools such as creating Firewall ACL rules

Help:

View basic help, manuals of command line tools, etc..

Wireshark Filter Toolbar

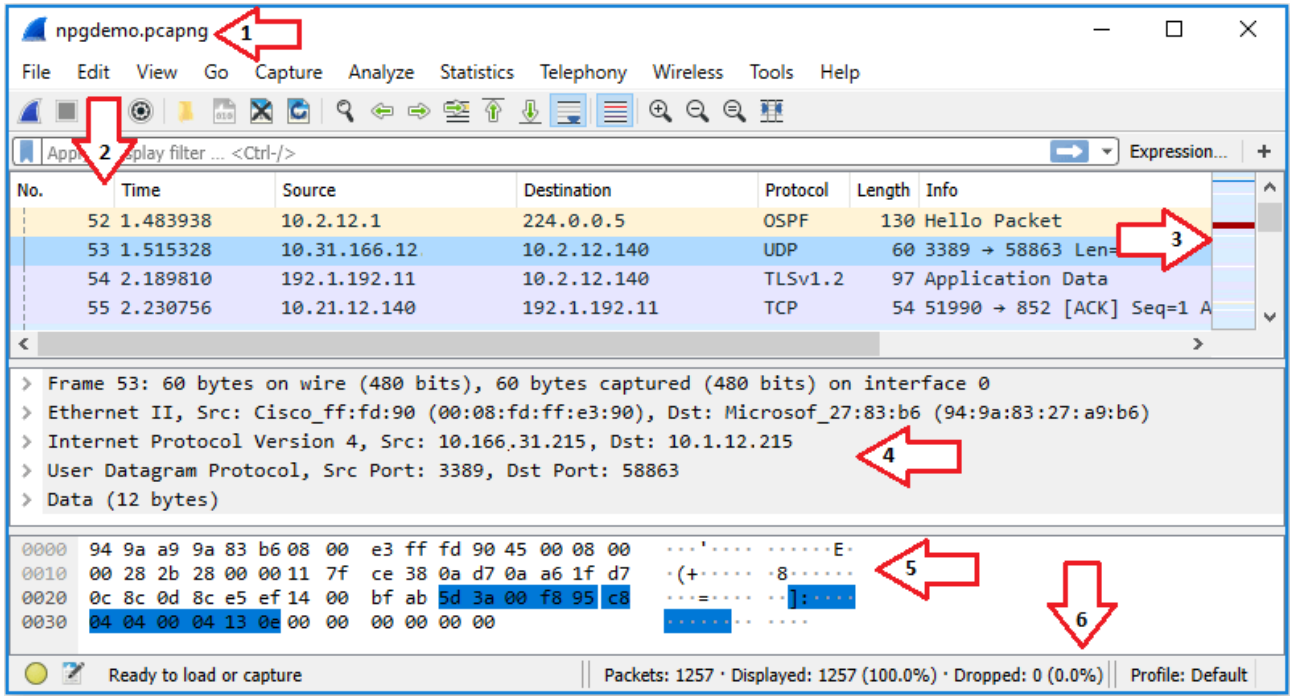
This toolbar allows you to quickly edit and apply display filters to your capture. Display filters allow you to narrow down the packets that you've captured to only those that are relevant to what you're trying to see such as specific IP address sources and destinations, protocols, MAC addresses, etc...



Wireshark Interface List

The Interface List is the area where the interfaces that your device has installed will appear. Before you can see packet data you need to pick one of the interfaces by clicking on it. You can choose a capture filter and type of interface to show in the interfaces lists at this screen as well.

Clicking on and interface or opening an existing capture file will take you to the working screen:



Primary Areas of the Wireshark Working Screen:

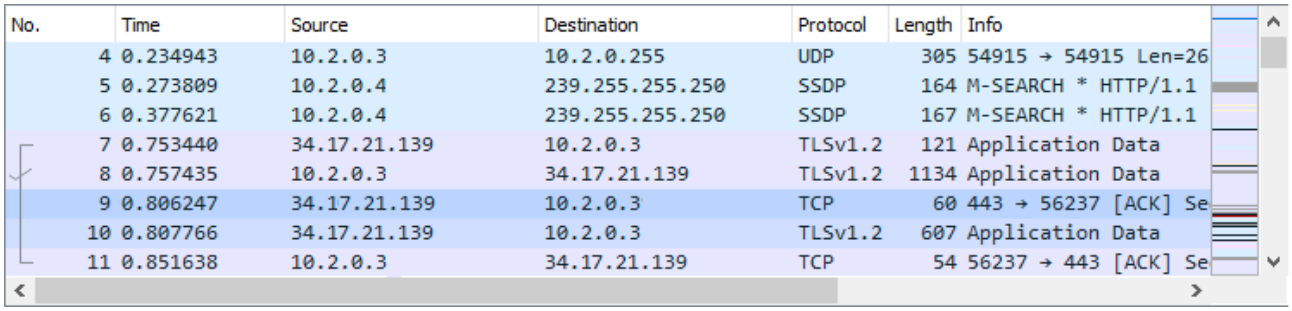
1. Title Bar
2. Packet List Pane
3. Intelligent Scrollbar
4. Packet Details Pane
5. Packet Bytes Pane

6. The Statusbar

1. Wireshark Title Bar:

This bar shows the name of the interface you're capturing until you save your capture. Then it will show the name of the capture dump file. If you open a saved capture file its' name will be displayed here.

2. Wireshark Packet List Pane:



No.	Time	Source	Destination	Protocol	Length	Info
4	0.234943	10.2.0.3	10.2.0.255	UDP	305	54915 → 54915 Len=26
5	0.273809	10.2.0.4	239.255.255.250	SSDP	164	M-SEARCH * HTTP/1.1
6	0.377621	10.2.0.4	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
7	0.753440	34.17.21.139	10.2.0.3	TLSv1.2	121	Application Data
8	0.757435	10.2.0.3	34.17.21.139	TLSv1.2	1134	Application Data
9	0.806247	34.17.21.139	10.2.0.3	TCP	60	443 → 56237 [ACK] Se
10	0.807766	34.17.21.139	10.2.0.3	TLSv1.2	607	Application Data
11	0.851638	10.2.0.3	34.17.21.139	TCP	54	56237 → 443 [ACK] Se

Every line in this pane represents one packet. By default, the pane is broken up into 7 columns, each of which provides useful identification data for each packet and can be sorted to help you better dissect the data. You can remove, add, and reorder the columns to suit your needs. Selecting a packet will show more details in the Packet Details Pane and Packet Bytes Pane.

- **No:**

The No. column assigns a unique number to each packet. It can also display a symbol to help identify the relationship between packets if you click on a packet.

- **Time:**

Displays the timestamp for when the packet was captured. The format of this timestamp is customizable.

- **Source:**

Displays the source IP or MAC address that the packet originated from.

- **Destination:**

Displays the destination IP or MAC address that the packet was heading to.

- **Protocol:**

Displays abbreviated protocol information for the packet.

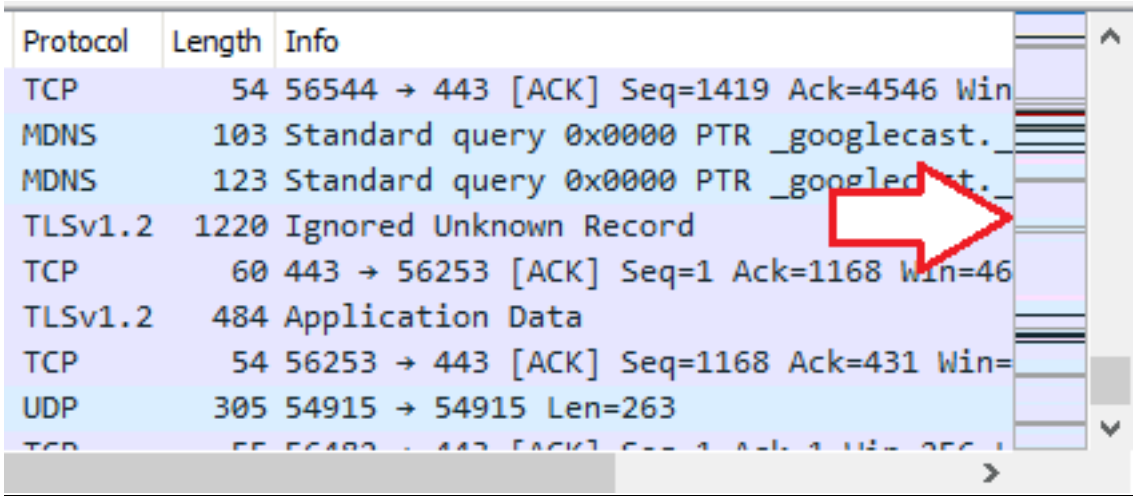
- **Length:**

Displays the packet length.

- **Info:**

Displays additional information related to the packet.

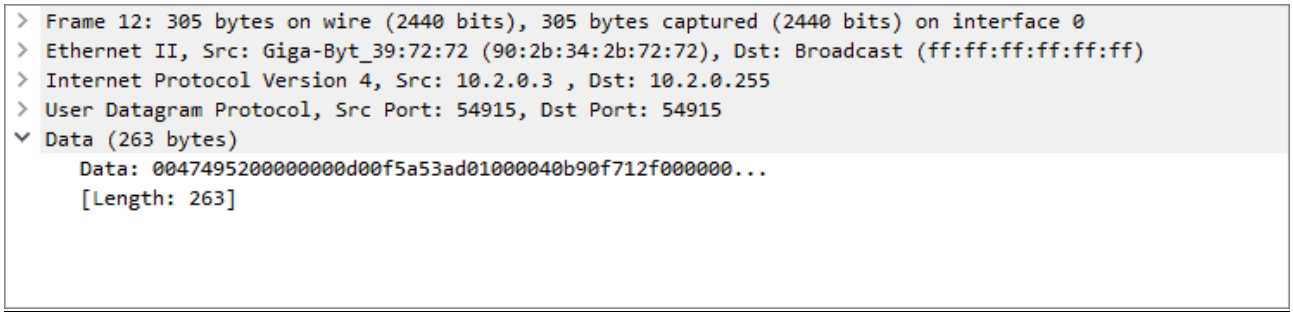
3. Wireshark Intelligent Scrollbar:



To the right of the Packet List Pane is a colored bar called the Intelligent Scrollbar which is a mini-map of packets. Every line of the Intelligent Scrollbar is a raster line that represents a single packet. The number of packets displayed depends on the height of the list and your physical display's specifications.

This list makes it easy to jump to packets based on colorization. It also makes it easy to spot groups of specific types of packets especially if you utilize custom coloring rules.

4. Wireshark Packet Details Pane:



When you click on a packet in the Packet List Pane it loads data about that packet in the Packet Details Pane. This pane displays the packet's different protocols and protocol fields. This is list is displayed as a tree that can be expanded to show even more detail.

The details can also include a couple special fields that Wireshark generates on its' own by analyzing the packets. The two fields are Generated Fields and Links**.**

- **Generated Fields:**

This information is enclosed in brackets ([ ]) and contains info such as TCP analysis, response time, checksum validation, and IP geolocation.

- **Links:**

Wireshark will generate a link if it detects relationships between packets. These links will be formatted blue with an underline. Double-clicking on the link will jump you to the related packet.

There is also a context menu which you can access by right clicking within the pane.

5. Wireshark Packet Bytes Pane:

```
0000  80 2a a8 5e 18 ad 90 2b  34 39 72 2d 08 00 45 00   ·*·^···+ 49r-··E·
0010  00 28 58 0e 40 00 80 06  60 49 0a 02 00 27 22 c5   ·(X·@··· `I···'"·
0020  15 8b db ad 01 bb 06 b9  68 32 52 75 f4 49 50 10   ···· ···· h2Ru·IP·
0030  00 fe 42 93 00 00                                  ··B···
```

When you click on a packet in the Packet List Pane it loads data about the packet in the Packet Bytes Pane as well. This data is in a hexdump style with each line displaying the data offset, 16 hexadecimal bytes, and 16 ASCII bytes. The periods represent non-printable bytes.

If you mouse over a specific piece of data Wireshark will highlight the corresponding data which you see in the example above in blue where the hex bytes are highlighted along with the associated ASCII bytes.

Occasionally, when Wireshark reassembles some packets into a single chunk or decrypts data, there will be multiple pages tabbed at the bottom of the Packet Bytes Pane.

There is also a context menu which you can access by right clicking within the pane.

6. Wireshark Statusbar:

The statusbar contains informational messages.

- **The Colorized Bullet:**

At the far left of the statusbar is a colorized bullet which represents the highest Expert Level. When moused over it displays the Expert Infos box.

- **The Capture File Properties Button:**

To the right of the colorized bullet is the capture file properties dialog button.

- **The Left Side:**

In most circumstances the left side will show contextual information about the capture file such as it's size, name, time elapsed. Hovering over the capture file details will show the path to the file.

- **The Middle:**

The middle will show info related to the current capture file such a the packet counts and load time.

- **The Right Side:**

The right will show the current configuration profile being used.

If you use a display filter that has a problem, it will also be displayed in the Capture File Details field of the statusbar.

Hopefully this Wireshark interface tutorial was easy to understand and has made Wireshark a lot easier for you to understand and work with. I recommend loading up Wireshark, picking an interface on your computer, and just poking around the interface and packets. You'll be amazed at the type of data you'll see crossing the wire.

6. Display Filters:

Display filters are used to show only specific packets instead of all captured packets

Format:

filter_name

Examples:

Filter	Purpose
http	HTTP packets only
dns	DNS packets only
tcp	TCP packets only

7. HTTP Filter:

Definition

HTTP (HyperText Transfer Protocol) is used for web browsing.

Filter Command

http

Result

Shows only:

- Web requests
- GET/POST requests
- Server responses

8. DNS Filter

Definition

DNS (Domain Name System) converts domain names into IP addresses.

Filter Command

dns

Result

Shows only:

- DNS queries
- DNS responses
- Domain lookups

9. TCP Filter:

Definition

TCP (Transmission Control Protocol) ensures reliable communication.

Filter Command

tcp

Result

Shows only:

- TCP handshake packets
- Data transmission packets
- Acknowledgements

10. Practical Procedure

Follow these steps during lab:

Step 1 – Start Capture

- Open Wireshark
- Select WiFi/Ethernet
- Click Start

Step 2 – Generate Traffic

- Open browser
- Visit any website (google.com, youtube.com)

Step 3 – Apply Filters

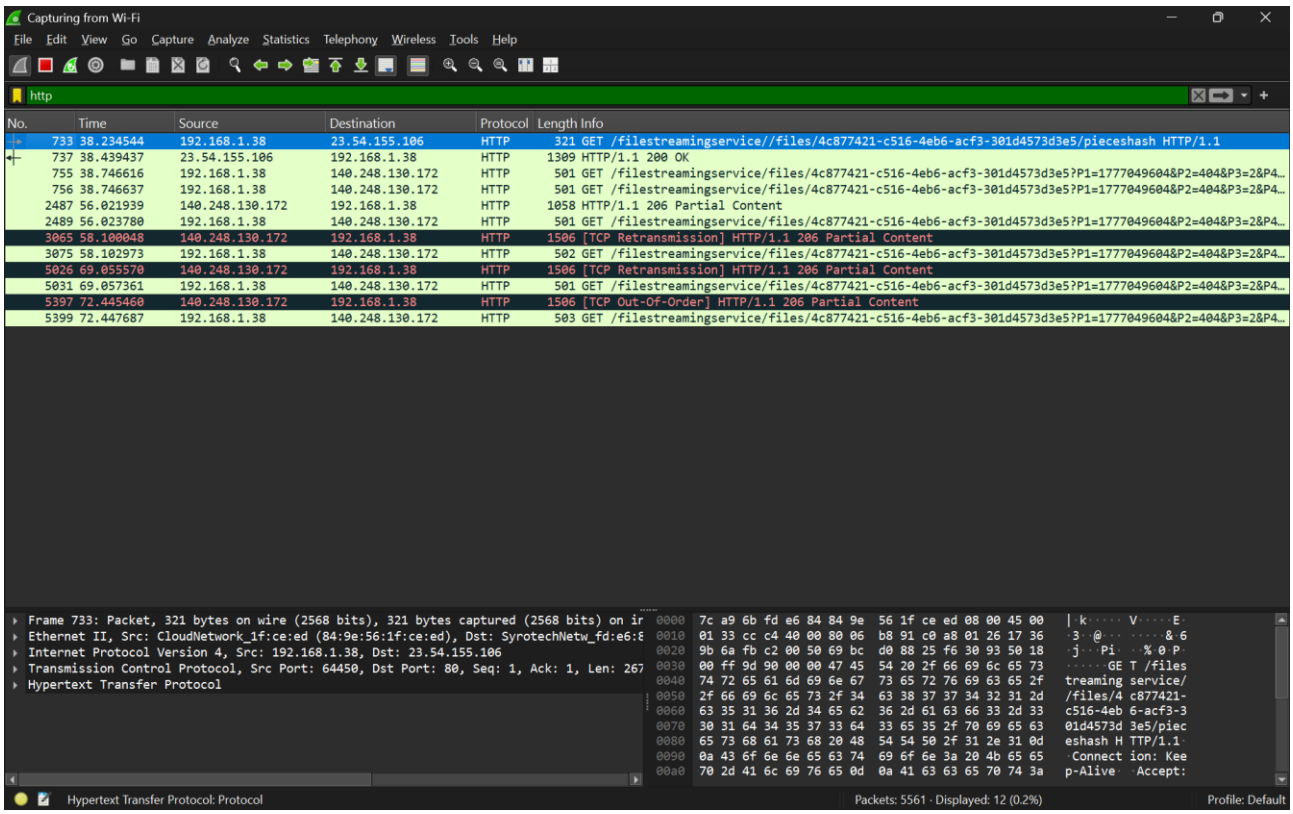
HTTP
Type:
http
Press "Enter"

DNS
Type:
dns
Press "Enter"

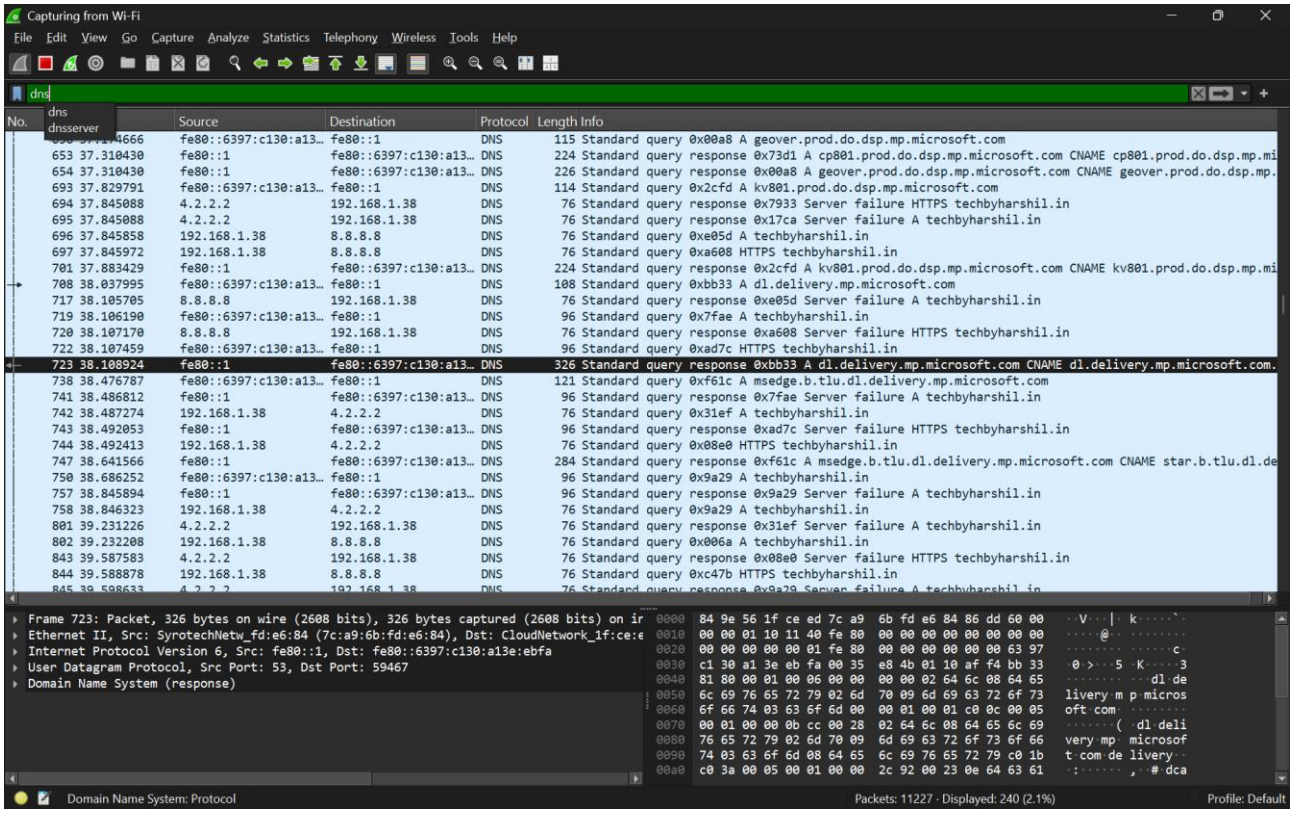
TCP
Type:
tcp
Press "Enter"

11. Output Screenshots

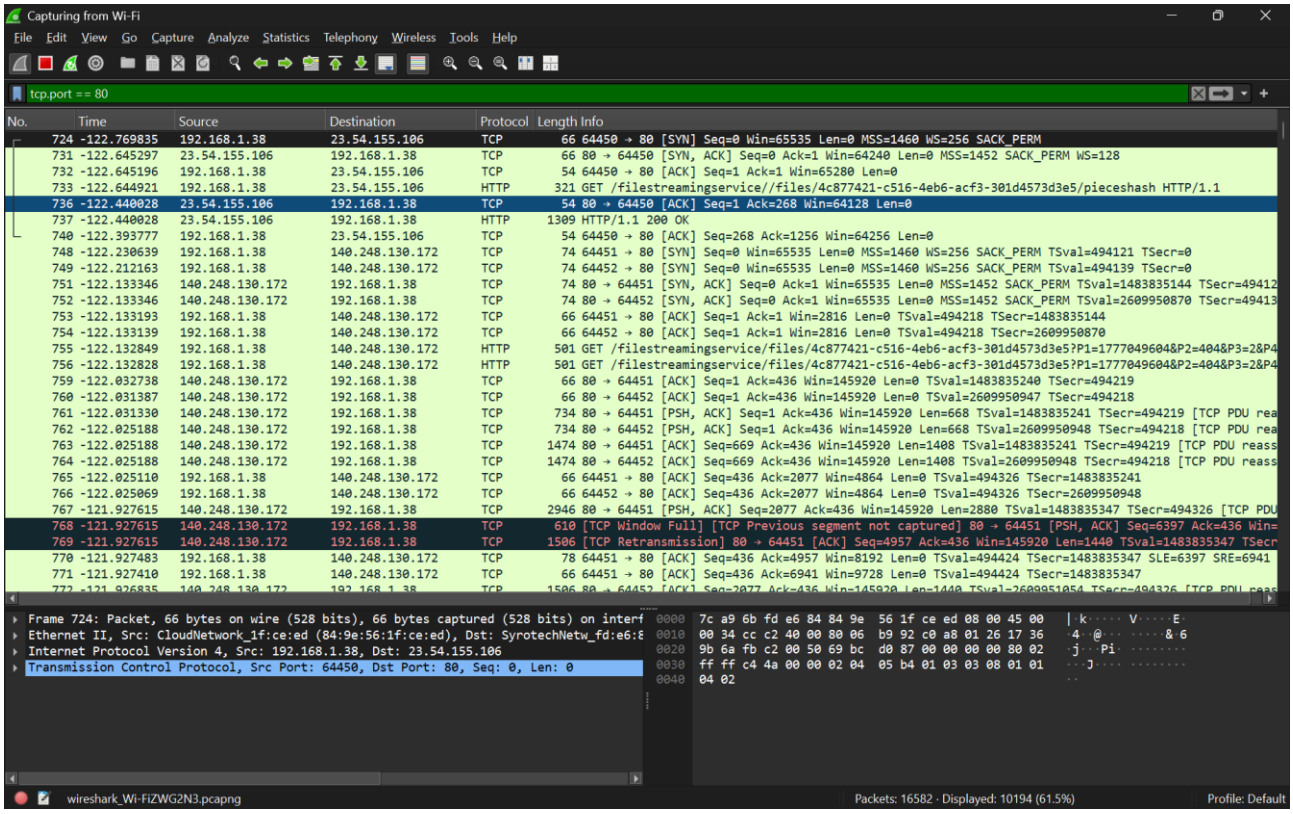
HTTP Filter Output



DNS Filter Output



TCP Filter Output



12. Conclusion

In this experiment, we successfully:

- Installed Wireshark
- Captured live packets
- Applied HTTP, DNS, and TCP filters
- Analyzed network traffic

We learned how Wireshark helps in network monitoring, troubleshooting, and protocol analysis. This tool is very useful for cybersecurity and networking professionals.

13. References:

- Wireshark Official Website - https://www.wireshark.org/
- Wireshark Documentation - https://networkproguide.com
- Research and analysis - ChatGPT

THANK YOU